

$$\textcircled{1} \quad \phi(a,b) = \sum_i d_i^2$$

$$\textcircled{2} \quad d_i = x_i - x(y_i)$$

$\Rightarrow$ first rearrange eqn of a line to be $x(y_i)$

$$y_i = ax_i + b$$

$$y_i - b = ax_i$$

$$x_i = \frac{y_i - b}{a}$$

$$\textcircled{3} \quad d_i = x_i - \frac{y_i - b}{a}$$

Plug $\textcircled{3}$ into $\textcircled{1}$:

$$\phi_{ab} = \sum_i \left[x_i - \left(\frac{y_i - b}{a} \right) \right]^2$$

$$\frac{\partial \phi_{ab}}{\partial a} = \sum_i \left[x_i - \frac{y_i - b}{a} \right] \left[a^{-2} (y_i - b) \right]$$

$$= \frac{-\sum_i (b - y_i)(b + ax_i - y_i)}{a^3}$$

$$\frac{\partial \phi_{ab}}{\partial b} = \sum_i \left[x_i - \left(\frac{y_i - b}{a} \right) \right] \left[\frac{1}{a} \right]$$

$$= \frac{\sum_i (b + ax_i - y_i)}{a^2}$$

Collect terms and turn into S_x , S_y , S_{yy} , S_{xx} and S_{xy}

$$\frac{\partial \phi_{ab}}{\partial a} = \frac{-2 [b^2 + bax_i - by_i - by_i - ax_i y_i + y_i^2]}{a^3}$$

$$= \frac{-2 [b^2 + bax_i - 2by_i - ax_i y_i + y_i^2]}{a^3}$$

$$= \frac{-2 [b^2 + baS_x - 2bS_y - aS_{xy} + S_{yy}]}{a^3}$$

$$\frac{\partial \phi_{ab}}{\partial b} = \frac{2(b + ax_i - y_i)}{a^2}$$

$$= \frac{2(b + aS_x - S_y)}{a^2}$$

$$\text{Solve } \frac{\partial \phi_{ab}}{\partial b} = 0 \text{ for } b \Rightarrow b = -aS_x + S_y$$

$$\frac{\partial \phi_{ab}}{\partial a} = 0$$

$$\frac{-2 [(-aS_x + S_y)^2 + (-aS_x + S_y)aS_x - 2(-aS_x + S_y)S_y - aS_{xy} + S_{yy}]}{a^3} = 0$$

$$\frac{-2 [-aS_{xy} + aS_x S_y - S_y^2 + S_{yy}]}{a^3} = 0$$

$$-2[-aS_{xy} + aS_xS_y - S_y^2 + S_{yy}] = 0$$

$$a(S_xS_y - S_{xy}) = S_y^2 - S_{yy}$$

$$a = \frac{(S_y^2 - S_{yy})}{(S_xS_y - S_{xy})}$$

$$b = -aS_x + S_y$$